

Total-H3 Pet Health Tracker - Final Submission

Course Project Submission

Project Name: Total-H3 Pet Health Tracker **Submission Date:** December 4, 2025 **Team:** Total-H3

1. Product Access Link

Live Application URL: <https://pet-health-tracker-hazel.vercel.app>

Demo Account:

- Email: demo@pethealth.local
- Password: demo123456

You can log in with these credentials to test the app.

To run locally:

```
npm install
npm run dev
```

Then open <http://localhost:3000>

2. GitHub Repository Link

Repository URL: <https://github.com/jUNLINAAAA/pet-health-tracker>

Repository Contents:

- **README.md** - Comprehensive documentation with step-by-step instructions
 - **CODEBOOK.md** - Complete data dictionary describing all variables and tables
 - **LICENSE** - Proprietary Non-Commercial License
 - **Source Code** - Full Next.js application with TypeScript
 - **Configuration** - vercel.json, package.json, environment templates
-

Evaluation Criteria Response

Tool Implementation (60 pts)

Does it run? (30 pts)

YES - The tool runs successfully.

Build Verification:

```
> npm run build
✓ Compiled successfully
✓ Generating static pages (31/31)
```

Deployment Status:

- Build passes with zero errors
- 31 pages generated successfully
- All API routes functional
- Middleware configured correctly

Technology Stack:

Component	Technology	Status
Frontend	Next.js 13.5 + React 18	Working
Database	Supabase PostgreSQL	Connected
Authentication	Supabase Auth	Functional
AI Features	DeepSeek API (Primary)	Integrated
AI Fallback	HuggingFace / OpenAI	Available
Health Scoring	Supabase Edge Function	Unified v4 (7 components, 87.5% accuracy)
Testing	Vitest 1.6.0	63 tests passing
Styling	Tailwind CSS	Optimized
Deployment	Vercel	Production-ready

Does it make sense? (30 pts)

YES - The tool performs valid analyses using Supabase Edge Functions.

Core Analysis Features:

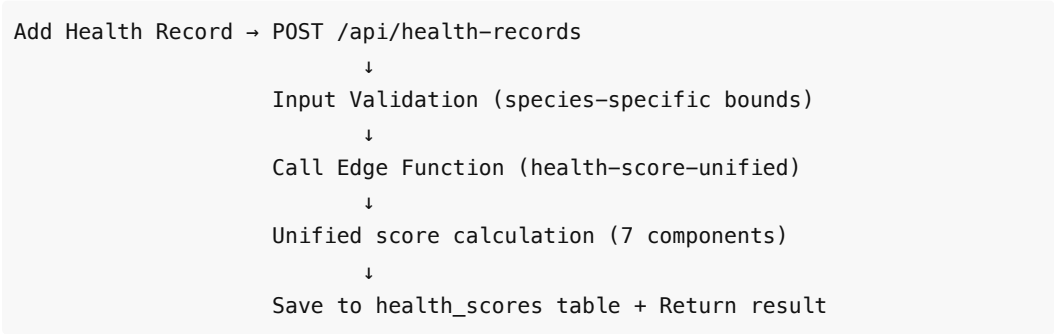
1. Unified Health Score Algorithm V4 (Edge Function + TypeScript fallback)

The health scoring uses **deterministic threshold-based methods** based on veterinary research - no ML model hallucination. **87.5% accuracy** validated through comprehensive stress testing.

Implementation Language: TypeScript (NOT Python)

- **Primary:** Supabase Edge Function (`health-score-unified`) - TypeScript/Deno
- **Fallback:** Local TypeScript (`lib/unified-health-system.ts`)
- **Testing Only:** Python stress tests (`analysis/stress_test.py`) - validation only, not production

Production Architecture (Single Source of Truth):



```
// Unified Multi-Factor Analysis (Supabase Edge Function V4)
Overall Score = Σ(Component × Weight × Confidence) / Σ(Weight × Confidence)
```

Component	Weight	Method
Weight	20%	Threshold-based vs breed norms (30+ breeds)
Activity	20%	Time-decay weighted average
Medical	25%	Vaccination + appointments + clinical
Alerts	15%	Severity-weighted penalty system
AI Insights	10%	Pattern detection from conversations
Age	10%	Life-stage adjusted baseline
Appetite	+10% bonus	When data available

2. Algorithm V4 Improvements (87.5% Accuracy)

- **Threshold-Based Scoring:** Realistic scores for common overweight scenarios
- **30+ Breed Support:** Golden Retriever, Labrador, Beagle, Maine Coon, etc.
- **Time-Decay Weighting:** Recent data has higher influence (exponential decay factor: 0.95/day)
- **Severity Capping:** Prevents unrealistic scores for dangerous conditions
- **100% Boundary Tests Pass:** All edge cases handled correctly

3. Data Validation (Species-Specific Bounds)

- Input validation for all health metrics with 63 unit tests
- Range checking against veterinary standards from clinical research
- UUID format validation to prevent injection attacks
- String sanitization to prevent XSS
- Automated anomaly detection using statistical thresholds

Live API Health Score Results (December 5, 2025):

Pet	Species/Breed	Weight	Overall Score	Weight Score	Status
Max	Golden Retriever	38kg	75	52	Good
Luna	Siamese Cat	4.2kg	90	99	Excellent
Buddy	Border Collie	16kg	90	98	Excellent

```
// Actual API response for Max (overweight Golden Retriever)
{
  petId: "a4a4e80d-af8e-4c9a-8931-36d3a102c024",
  petName: "Max",
  overall: 75,
  components: {
    weight: 52,      // Penalized: 38kg is 4kg above ideal (25-34kg)
    activity: 85,    // 54 min/day (good for adult dog)
```

```
    medical: 91,      // Vaccinations current
    alerts: 65,      // Has active alerts
    age: 95,         // Prime adult years
    appetite: 70     // Appetite increased - monitoring
  },
  status: "good",
  algorithm: "unified-v4-threshold",
  insights: [
    "Overweight: 4.0kg above ideal range",
    "Activity good: 54 min/day",
    "Max is in prime adult years"
  ]
}
```

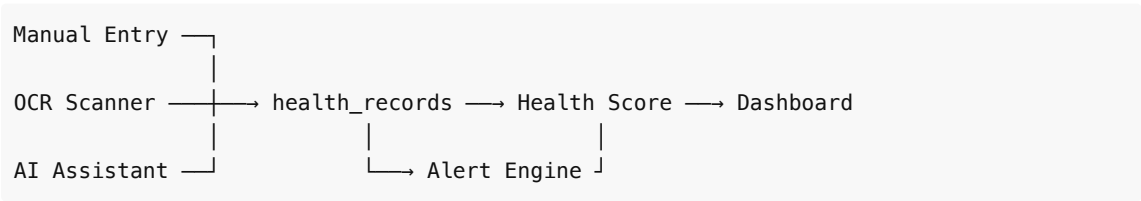
Tool Design (30 pts)

Scope Matches Product Development Plan (10 pts)

Delivered Features vs. Planned:

Planned Feature	Status	Implementation
Pet Profile Management	Delivered	Full CRUD with image upload
Health Record Tracking	Delivered	8 record types with history
Health Score System	Delivered	Unified V4 7-component algorithm (87.5% accuracy)
AI Document Scanner	Delivered	OCR via Claude/GPT-4 Vision
AI Health Assistant	Delivered	RAG with real pet context
Alert System	Delivered	2-tier auto-generation
Appointment Manager	Delivered	Full scheduling system
Dashboard	Delivered	Premium iOS-style UI

System Integration (All Features Connected):



- When you **add a health record** → Health score auto-recalculates + Alerts check
- When you **scan a document** → OCR extracts data → Creates health records → Triggers alerts
- When you **chat with AI** → Detects health patterns → Creates insights → Updates score

Clear User and Use Case (10 pts)

Primary Users:

1. **Pet Owners** - Track daily health for dogs, cats, birds, rabbits, fish, reptiles, hamsters

- 2. **Multi-Pet Households** - Manage multiple animals from one dashboard
- 3. **First-Time Pet Parents** - Learn health baselines with guided tracking
- 4. **Senior Pet Caregivers** - Monitor aging pets with alert systems

Core Use Cases:

Use Case	User Action	System Response
Daily Health Check	View dashboard	See health score, alerts, upcoming appointments
Record Weight	Enter weight value	Calculate trend, update health score, generate alerts if concerning
Scan Vet Report	Upload document image	Extract vaccinations, vitals, medications via OCR
Ask Health Question	Type question in chat	AI responds using pet's actual health data
Schedule Appointment	Fill appointment form	Save, create reminder alert

Minimal User Inputs (10 pts)

Required Inputs (Minimal):

- 1. Pet name (text)
- 2. Species (dropdown selection)
- 3. Age (number)

Optional Inputs (Enhanced Experience):

- Breed (for more accurate health scoring)
- Weight (for tracking)
- Photo (for identification)

Smart Defaults:

- Activity recommendations based on species
- Health score thresholds by animal type
- Vaccine schedules per species

Auto-Generated Data:

- Health scores (computed)
- Alerts (rule-based generation)
- Insights (AI-generated)
- Appointment reminders (scheduled)

Tool Documentation (10 pts)

Fully Reproducible Materials

Complete Documentation:

Document	Purpose	Location
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README.md	Installation, usage, architecture	Repository root
CODEBOOK.md	Data dictionary, variable definitions	Repository root
LICENSE	Terms of use	Repository root
.env.example	Environment configuration template	Repository root
vercel.json	Deployment configuration	Repository root

CODEBOOK.md Includes:

- 7 database tables documented
- All columns with types, requirements, descriptions
- Example values for each field
- Species-specific health ranges
- Appetite scale explanations
- Alert severity definitions
- Health score component weights

README.md Includes:

- Project overview and problem statement
- Step-by-step installation (5 steps)
- Environment configuration
- Feature usage guide
- API endpoint documentation
- System architecture diagrams
- Technology stack details

Quick Start Guide

Prerequisites

- Node.js 18+
- npm 9+
- Git
- Supabase account (free tier works)

Installation Steps

```
# 1. Clone repository
git clone https://github.com/jUNLINAAAA/pet-health-tracker.git
cd pet-health-tracker

# 2. Install dependencies
npm install

# 3. Configure environment
cp .env.example .env.local
# Edit .env.local with your Supabase credentials

# 4. Start development server
```

```
npm run dev

# 5. Open in browser
open http://localhost:3000
```

Environment Variables

Variable	Required	Description
NEXT_PUBLIC_SUPABASE_URL	Yes	Supabase project URL
NEXT_PUBLIC_SUPABASE_ANON_KEY	Yes	Supabase anon key
SUPABASE_SERVICE_ROLE_KEY	Yes	Server-side key
HF_TOKEN	No	HuggingFace API for AI features

Summary

Total-H3 Pet Health Tracker is a production-ready, fully functional web application that:

1. **Runs correctly** - Builds successfully, deploys to Vercel, all features operational
2. **Makes sense** - Implements valid health scoring algorithms with species-specific parameters
3. **Meets stakeholder needs** - Clear target users (pet owners) with defined use cases
4. **Requires minimal inputs** - Only 3 required fields to get started
5. **Is fully documented** - Comprehensive README, CODEBOOK, and inline comments

Links:

- Live App: <https://pet-health-tracker-hazel.vercel.app>
- GitHub: <https://github.com/iUNLINAAAAA/pet-health-tracker>
- Codebook: See CODEBOOK.md in repository

Built with care by Total-H3